

Operator Splitting

Goal: solve a composite problem with an iterative procedure that involves operators that deal with each term separately.

Ex: proximal gradient alg: $\min: f(x) = g(x) + h(x)$

Solve: $0 \in F(x)$, for $F = A + B$

F maximal monotone : $0 \in F(x)$.

$F = A + B$, where A, B max. monotone.

Prop 1: $0 \in A(x) + B(x) \Leftrightarrow \begin{cases} C_A \circ C_B(z) = x \\ x = R_B(z) \end{cases}$

Recall: Resolvent of A , $\lambda \geq 0$: $R_A = (I + \lambda A)^{-1}$

Calydely operator: $C_A = 2R_A - I$

Pf: $C_A C_B(z) = x, x = R_B(z) \Leftrightarrow \{ \overset{①}{x} = R_B(z), \overset{②}{\tilde{z}} = 2x - z, \overset{③}{\tilde{x}} = R_A(\tilde{z}), \overset{④}{z} = 2\tilde{x} - \tilde{z} \}$

③ - ④ : $\tilde{z} - z = 2x - z - (2\tilde{x} - \tilde{z}) = 2(x - \tilde{x}) + \tilde{z} - z \Leftrightarrow \boxed{x = \tilde{x}}$

④ : $2x = \tilde{z} + z$.

① : $x = R_B(z) \Leftrightarrow x = (I + \lambda B)^{-1} z \Leftrightarrow z \in x + \lambda B(x)$ (i)

③ : $x = \tilde{x} = R_A(\tilde{z}) \Leftrightarrow \tilde{z} \in x + \lambda A(x)$ (ii)

(i) + (ii) $2x = z + \tilde{z} \in 2x + \lambda(A(x) + B(x)) \Rightarrow 0 \in A(x) + B(x)$

Peaceman-Rachford, and Douglas-Rachford Splitting

To solve: $0 \in A(x) + B(x)$

① $\boxed{z_{t+1} \leftarrow C_A \circ C_B(z_t)}$ Peaceman-Rachford.

C_A, C_B non-expansive $\Rightarrow C_A \circ C_B$ non-expansive,

Douglas-Rachford Splitting - damped PR

$$z_{t+1} \leftarrow \underbrace{\left(\frac{1}{2} I + \frac{1}{2} C_A \circ C_B \right)}_{\text{firmly non-expansive}} z_t$$

Thus, from our work on α -averaged operators
if $0 \in A(x) + B(x)$ has a solution, then $\text{Fix}[C_A \circ C_B] \neq \emptyset$
and $z_t \rightarrow \text{Fix}[C_A \circ C_B]$

Applying to — min: $f(x) + g(x)$

$$\text{DR: } \left[\begin{array}{l} z_{t+1} = \frac{1}{2} (I + C_A C_B) z_t, \quad \lambda = R_B(z) \end{array} \right]$$

$$R_A = (I + \lambda A)^{-1}, \quad C_A = 2R_A - I$$

For us: $A = \partial f$, $B = \partial g$

$$R_A(z) = \text{prox}_f(z), \quad R_B(z) = \text{prox}_g(z)$$

$$\text{DR: } \left[\begin{array}{l} z_{k+1} = \text{prox}_f(y_k) \quad x \\ y_{k+1} = y_k + \text{prox}_g(2x_{k+1} - y_k) - x_{k+1} \end{array} \right]$$

$$x_{k+1} = (I - \partial f)^{-1}(y_k) = R_f(y_k)$$

$$y_{k+1} = y_k + R_g(\underbrace{(2R_f - I)}_{C_f} y_k) - R_f(y_k)$$

$$= y_k + R_g \circ C_f(y_k) - R_f(y_k)$$

$$= R_g C_f(y_k) - \frac{1}{2} (2R_f(y_k) - y_k) + \frac{1}{2} y_k = \downarrow R_g C_f(y_k) - \frac{1}{2} C_f(y_k) + \frac{1}{2} y_k$$

$$= \frac{1}{2} y_k + \frac{1}{2} (2R_g C_f(y_k) - C_f(y_k)) = \frac{1}{2} (I + C_g C_f)(y_k).$$

Equivalent Formulations of DR

$$x_{k+1} = \text{prox}_f(y_k)$$

$$y_{k+1} = y_k + \text{prox}_g(2x_{k+1} - y_k) - x_{k+1}$$

$$y_{k+1} = y_k + \text{prox}_g(2x_k - y_k) - x_k$$

$$u_{k+1} = \text{prox}_f(y_{k+1})$$

$$u_{k+1} = \text{prox}_g(2x_k - y_k)$$

$$x_{k+1} = \text{prox}_f(y_k + u_{k+1} - x_k)$$

$$y_{k+1} = y_k + u_{k+1} - x_k$$

$$(w_k = x_k - y_k)$$

$$u_{k+1} = \text{prox}_g(x_k + w_k)$$

$$x_{k+1} = \text{prox}_f(u_{k+1} - w_k)$$

$$w_{k+1} = w_k + x_{k+1} - u_{k+1}$$

Scaling

$$\min: \underbrace{\lambda f(x)} + \underbrace{\lambda g(x)}$$

$$\text{DR: } x_{k+1} = \text{prox}_{\lambda f}(y_k)$$

$$y_{k+1} = y_k + \text{prox}_{\lambda g}(2x_{k+1} - y_k) - x_{k+1}$$



$$u_{k+1} = \text{prox}_{\lambda g}(x_k + w_k)$$

$$x_{k+1} = \text{prox}_{\lambda f}(u_{k+1} - w_k)$$

$$w_{k+1} = w_k + x_{k+1} - u_{k+1}$$